Qingkai Dong

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Professional Summary

Statistics Ph.D. student, expected to graduate in May 2027, with Servier Pharmaceuticals experience in individual patient data reconstruction and deep learning methods for interim analysis. Develops rare-feature subsampling methods, simulation studies, and reproducible software in R and Python.

Education

University of Connecticut Aug 2023 – Expected May 2027
Ph.D. in Statistics Storrs, CT, USA
Advisors: Prof. HaiYing Wang and Prof. Jun Yan

Zhongnan University of Economics and Law Sep 2020 – May 2023
MS in Mathematical Statistics Wuhan, China

Qingdao University Sep 2016 – May 2020
BS in Applied Statistics Qingdao, China

Technical Skills

Statistical Methods: survival analysis; interim analysis & conditional power; design simulation; big-data subsampling; model averaging; variable selection

Programming: R (package development), Python (machine learning / deep learning), SAS, Git, LaTeX

Industry and Research Experience

Global Biometrics Intern Jun 2026 – Aug 2026
Servier Pharmaceuticals

- Built a modular workflow to reconstruct synthetic individual patient data from published Kaplan–Meier curves and summary statistics using target-matching loss functions.
- Demonstrated close agreement with reported survival metrics and evaluated accuracy for additional survival quantities in simulation studies; applied the workflow to internal data.

Research Assistant (Academia–Industry Collaboration) Mar 2024 – Dec 2025
Servier Pharmaceuticals and the University of Connecticut

- Developed and implemented a covariate-informed deep learning framework to predict event times for censored participants and augment interim survival data for conditional-power estimation and futility decisions.
- Designed evaluation metrics for conditional-power forecast accuracy and futility decision quality. Conducted simulation studies to characterize when performance improves or degrades under varied censoring, covariate informativeness/type, and prediction range.
- Co-first-authored the methodology manuscript, under revision at *Statistics in Medicine*.

Research Assistant (Statistical Methodology) Aug 2023 – Present
University of Connecticut

- Developed rare-feature-aware subsampling methods for regression to preserve representation of rare binary covariates. Implemented the methods in the subsampling R package.
- Established theory linking the efficiency of rare-covariate coefficient estimation to the number of rare observations, explaining numerical instability and slow convergence.
- Evaluated the methods through simulation studies and applications to rare disease data. First-authored a methodology manuscript (under review at *JASA*).

Research Assistant (Breast Cancer Survivorship) Jul 2024 – Aug 2024
Departments of Statistics and Human Development and Family Sciences, University of Connecticut

- Retrieved, cleaned, and integrated All of Us EHR, survey, and physical-measurement data to prepare inputs for subsequent statistical modeling.
- Contributed to study framing on social determinants of health, frailty, and accelerated aging among breast cancer survivors.

Cross-Disciplinary Collaboration (Chemical Sensor Data Analytics) Mar 2024 – May 2024
Departments of Statistics and Chemistry, University of Connecticut

- Applied clustering and statistical analysis to porphyrinoid-sensor fluorescence data to assess compound differentiation and inform sensor selection.
- Produced interpretable summaries and R visualizations to support collaborators' experimental decisions.

Cross-Disciplinary Collaboration 2025 – Present
University of Connecticut & collaborators

- Coauthored a survey on knowledge localization in large language models, distinguishing representational, causal, and editable localization and synthesizing evidence and measurement practices.
- Coauthored a survey on time-series Explainable AI (XAI), organizing methods into signal, pattern, structural, and semantic explanations and reviewing LLM-enabled semantic explanations and benchmark resources.

Software

R package subsampling (CRAN)

- Implemented sampling-probability computation, pilot sampling, and two-step designs for GLMs, quantile regression, softmax regression, and rare-event/rare-feature settings.
- Delivered a modular R API with documentation, vignettes, and reproducible examples.

Selected Papers

Dong Q, Yan J, Zhang Q, Wang H. Rare-feature-adaptive subsampling for big data regression. Under review at *JASA*.
Wang D[†], **Dong Q**[†], Zhang Q, Hu Q, Wang H. Random Subsampling: Inferential Targets, Estimators, and Sampling Procedures. *Under review.* [†]Equal contribution.
Dong Q[†], Man A[†], Hua Z, Teng Z, Wang H, Yan J. Predictive-Modeling–Assisted Interim Decision-Making in Adaptive Trials with Censored Survival Outcomes. Under revision at *Statistics in Medicine.* [†]Equal contribution.
Dong Q, Liu B, Zhao H. Weighted Least Squares Model Averaging for Accelerated Failure Time Models. *Computational Statistics and Data Analysis*, 2023.
Zhao H, **Dong Q**. Variable Selection for Additive Hazards Model with Current Status Data. *Journal of Systems Science and Mathematical Sciences*, 2022.
Zhao H, Liu B, **Dong Q**. Jackknife Model Averaging of AFT Model with Current Status Data. *Acta Mathematicae Applicatae Sinica*, 2023.
Chen Z, Lucchesi G, **Dong Q**, Xu Z, Zheng X, Chen Y, Sha M, Cheng W, Ni J, Luo D. Where Does Knowledge Live? Representational, Causal, and Editable Localization in Language Models: A Survey. Preprint under review at TMLR, 2026.
Chen Z, Lucchesi G, **Dong Q**, Zheng X, Song D, Wen Q, Cheng W, Ni J, Luo D. From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens. Manuscript under review at *ACM Computing Surveys*, 2025.

Teaching Experience

University of Connecticut
Principal Instructor (Upcoming Appointment) Spring 2027
Scheduled to teach STAT 3255: Introduction to Data Science.

Teaching Assistant Sep 2026 – Dec 2026
Supporting STAT 1100Q: Elementary Concepts of Statistics and STAT 3445: Introduction to Mathematical Statistics II.

Principal Instructor Jan 2026 – May 2026
STAT 3675Q: Statistical Computing.
Taught R/RStudio for data management, visualization, reproducible analysis, and statistical modeling.

Academic Presentations

Invited talks (2026): Joint Statistical Meetings (JSM); New England Statistics Symposium (NESS); ICSA Applied Statistics Symposium; SILVER Online Lab Meeting, Department of Epidemiology & Public Health, University of Maryland School of Medicine.

Posters: New England Rare Disease Statistics Workshop (NERDS, 2025, Oct 2026); NESS (2026); The Design and Analysis of Experiments (DAE, 2026); Dahshu Data Science Symposium (2025).

Academic Service

Journal Reviewer: *Computational Statistics and Data Analysis*; *Journal of Computational Science*; *Statistical Papers*; *Sankhya B*; *Journal of Systems Science and Mathematical Sciences (Chinese)*

Awards

Student Poster Award, New England Rare Disease Statistics Workshop (NERDS) Oct 2026
UConn Graduate School Conference Participation Award Jul 2026
New England Statistics Symposium (NESS) Poster Award 2026
ICSA Applied Statistics Symposium Student Paper Award 2026
Predoc Fellowship - University of Connecticut 2025
First-class Scholarship - Zhongnan University of Economics and Law 2020, 2022